Microsoft Azure Fundamentals (AZ-900) — Section 2 Lab

Cloud Computing Concepts: Service Models, Pricing & Cost Management

> **Lab Provider:** Dion Training | **Estimated Duration:** 15 minutes
> **Platform:** Microsoft Azure (Free Account) | **Tools Used:** Azure Pricing Calculator · Azure Portal · Cost Management + Billing **Completed by:** **Oluwaseun Osunsola**

📋 Lab Overview

In this hands-on lab, I took on the role of IT consultant for **CloudFirst Retail**, a startup e-commerce company evaluating Azure deployment options. The objective was to compare IaaS vs. PaaS service models, estimate monthly costs, analyze SLA commitments, and explore Azure's consumption-based cost management tools — all in support of a data-driven cloud adoption recommendation.

🎯 Learning Objectives

- Compare IaaS and PaaS pricing using the Azure Pricing Calculator
- Identify SLA differences between service models
- Navigate the Azure Cost Management + Billing portal
- Understand consumption-based (OpEx) pricing through hands-on resource exploration
- Apply the Shared Responsibility Model to real service selection decisions

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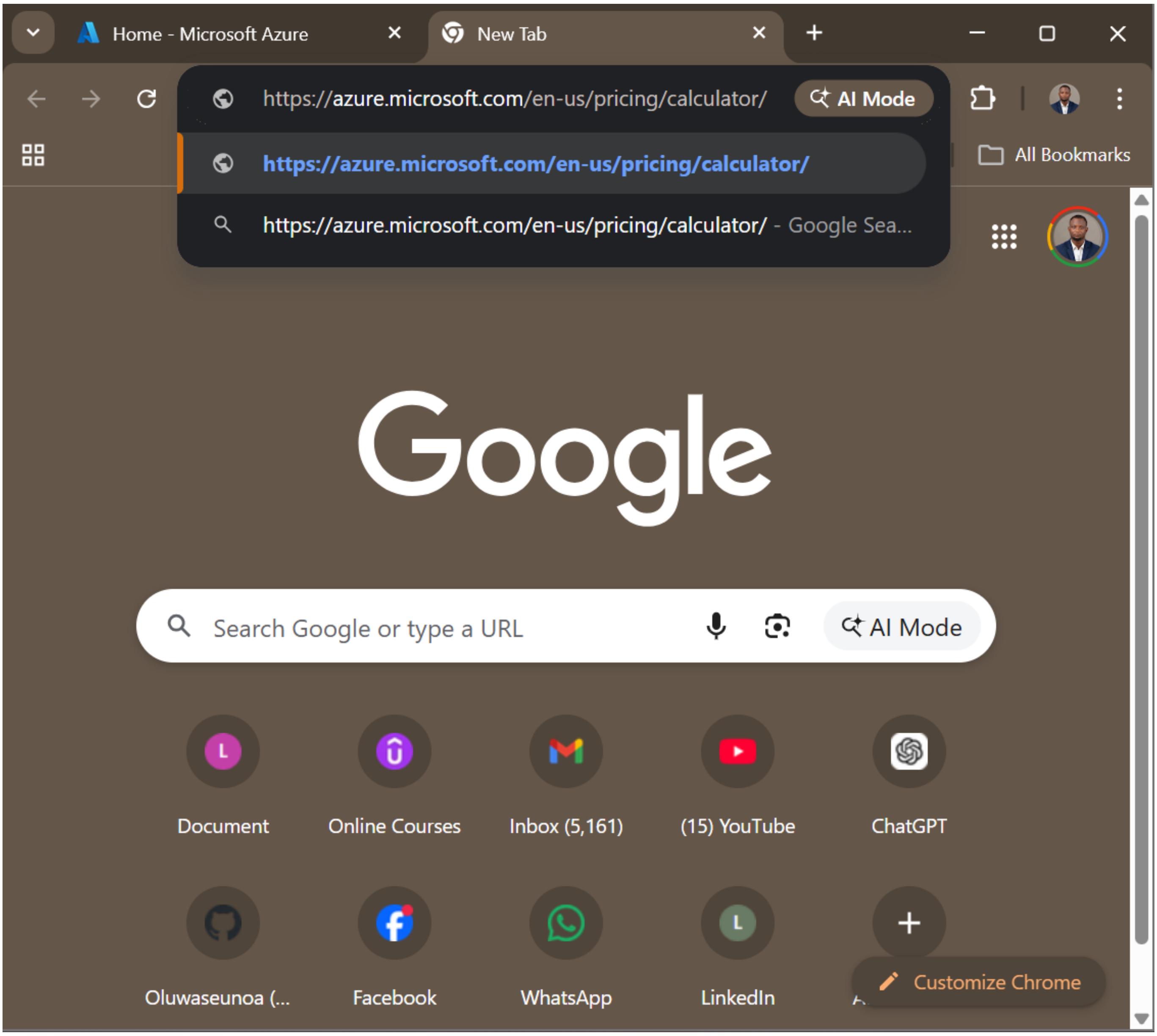
🧰 Prerequisites

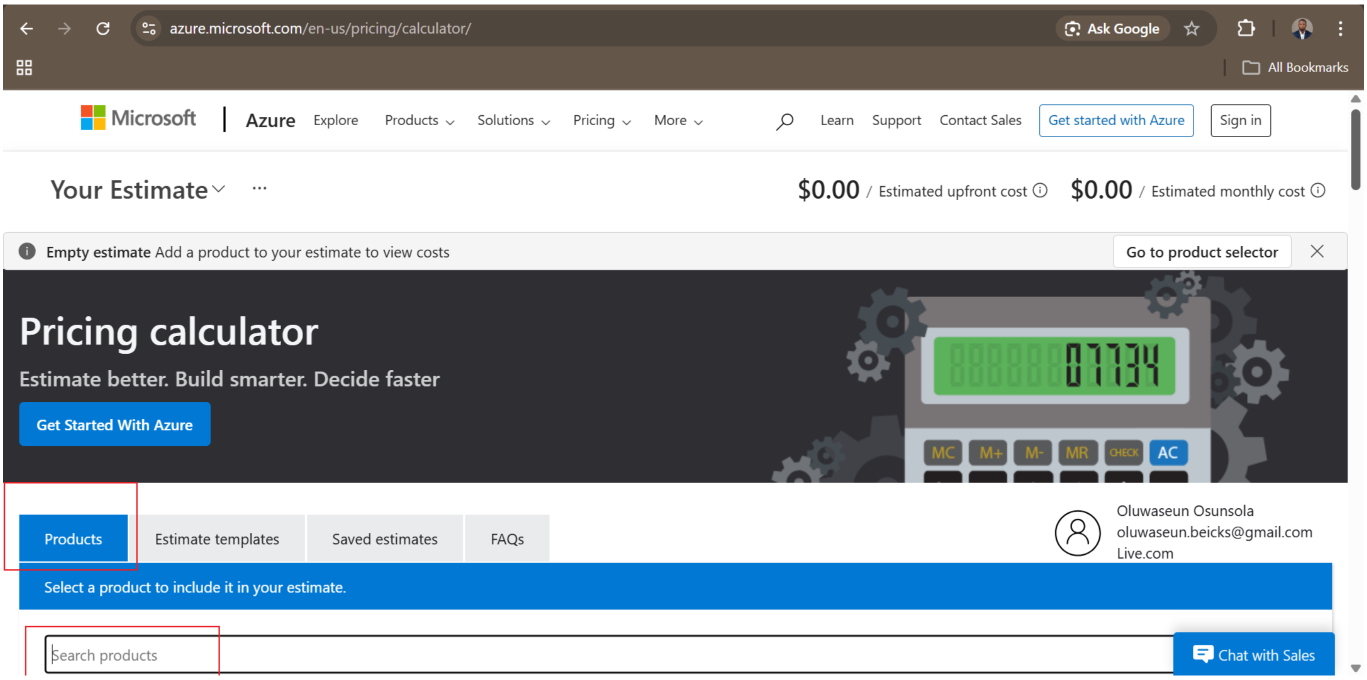
- Active Azure free account (no credit card charges)
- Web browser access
- 10–15 minutes of focused time

🪜 Step-by-Step Walkthrough

Step 1 — Launch the Azure Pricing Calculator

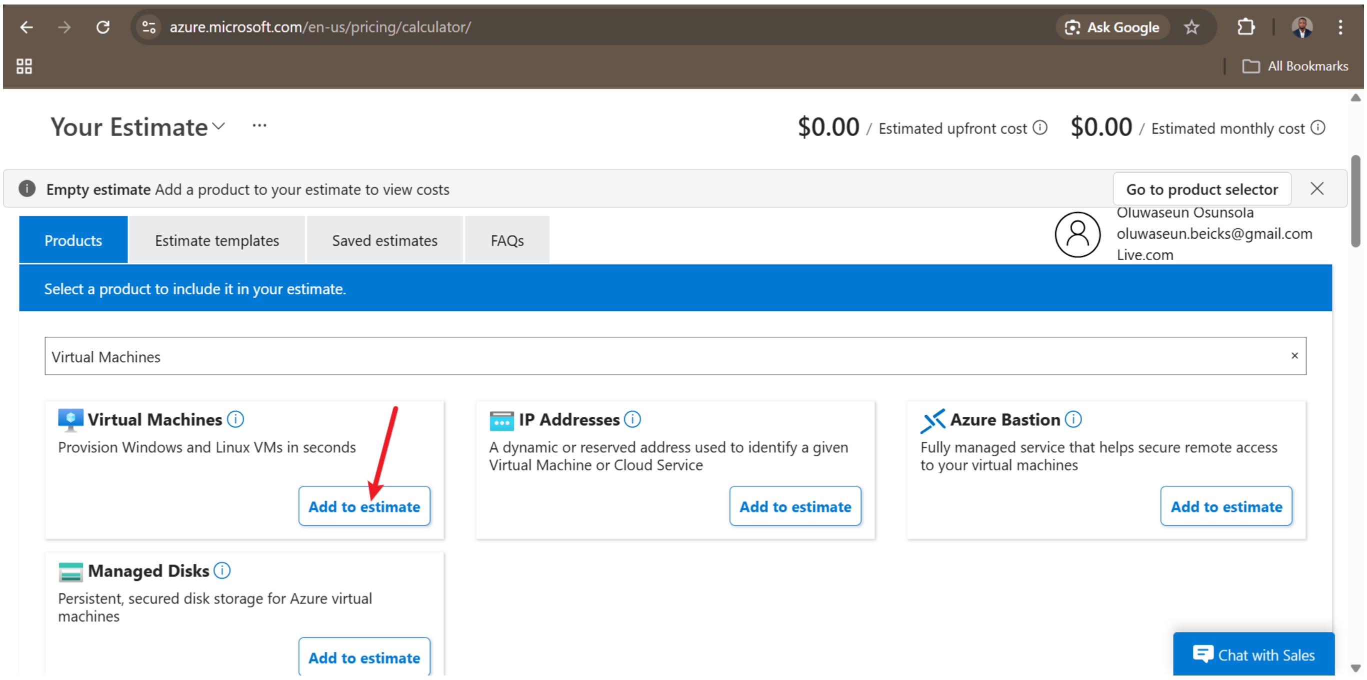
Navigated to the Azure Pricing Calculator at `https://azure.microsoft.com/en-us/pricing/calculator/` to begin estimating costs before any resource deployment. This reinforces Azure's OpEx model and prevents budget surprises.





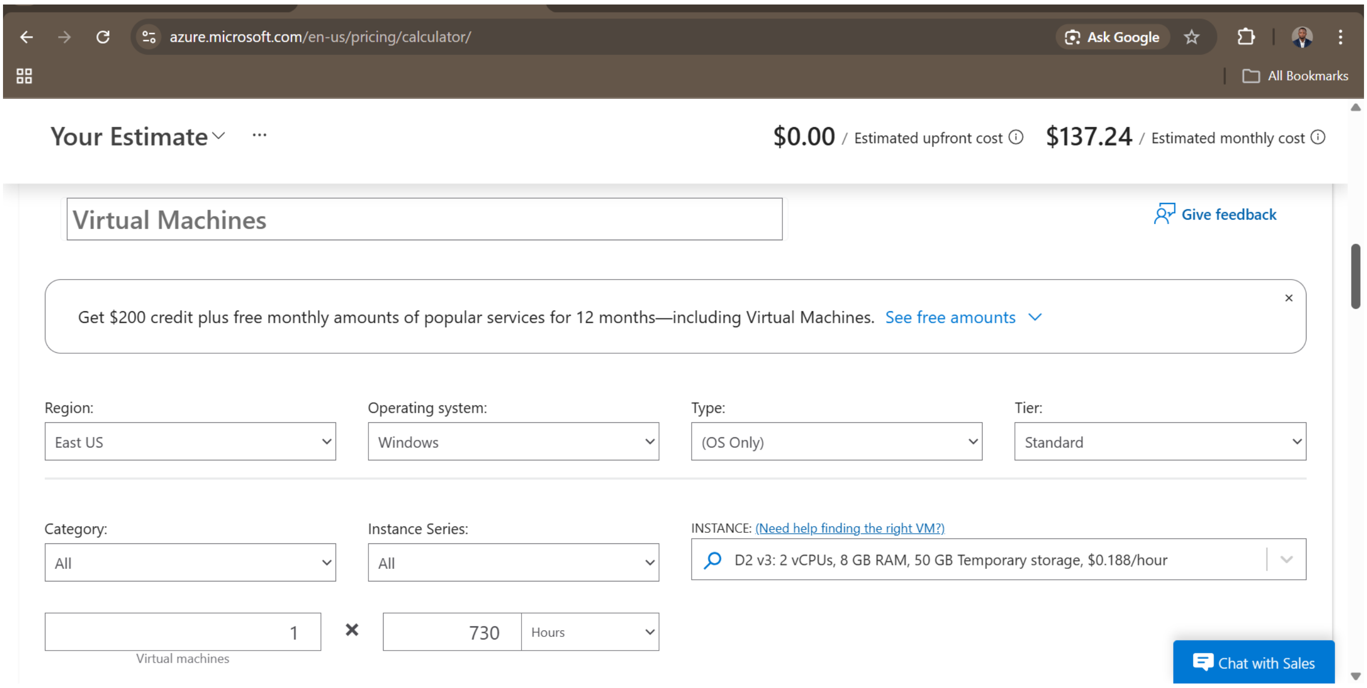
Step 2 — Estimate IaaS Costs (Virtual Machines)

Searched for **Virtual Machines** in the Products search bar and added it to the estimate. This demonstrates IaaS — maximum control, but also maximum customer responsibility.

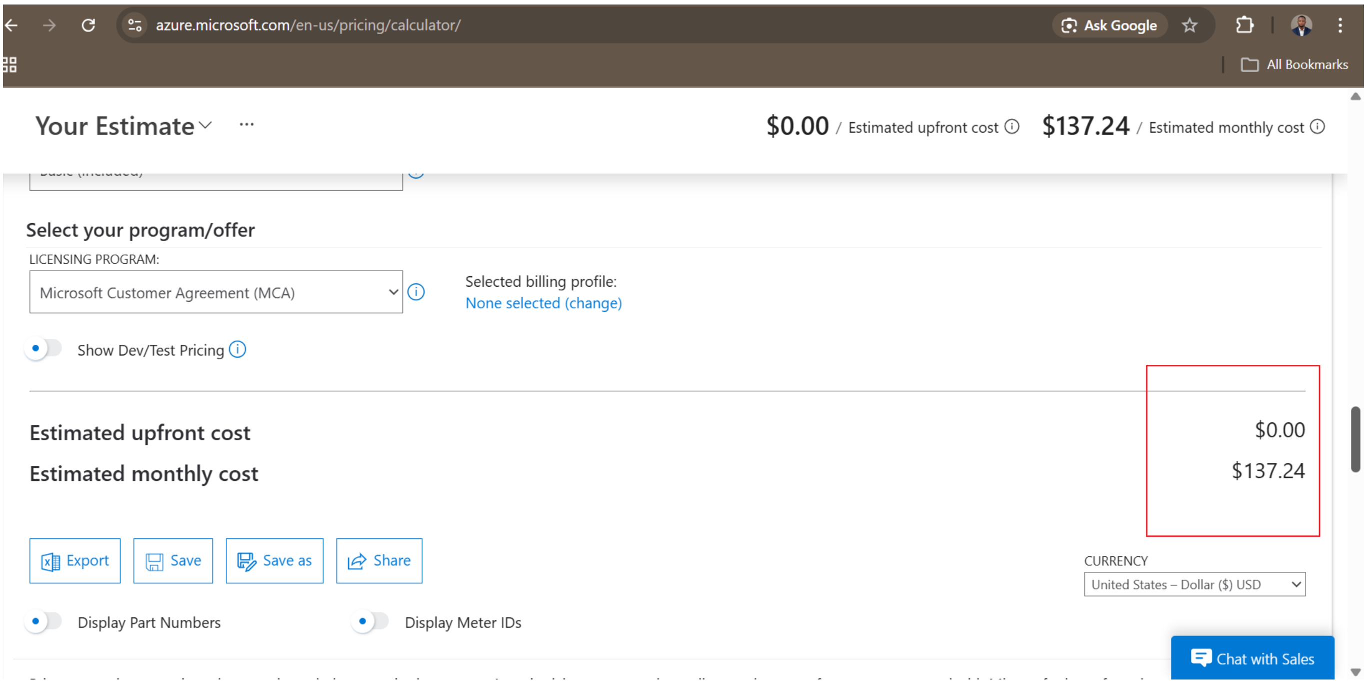


Scrolled down to configure the VM settings:

- **Region:** East US
- **OS:** Windows
- **Tier:** Standard
- **Instance:** D2s v3 (2 vCPUs, 8 GB RAM)
- **Count:** 2 VMs (for high availability)
- **Hours:** 730 (full month, 24/7)

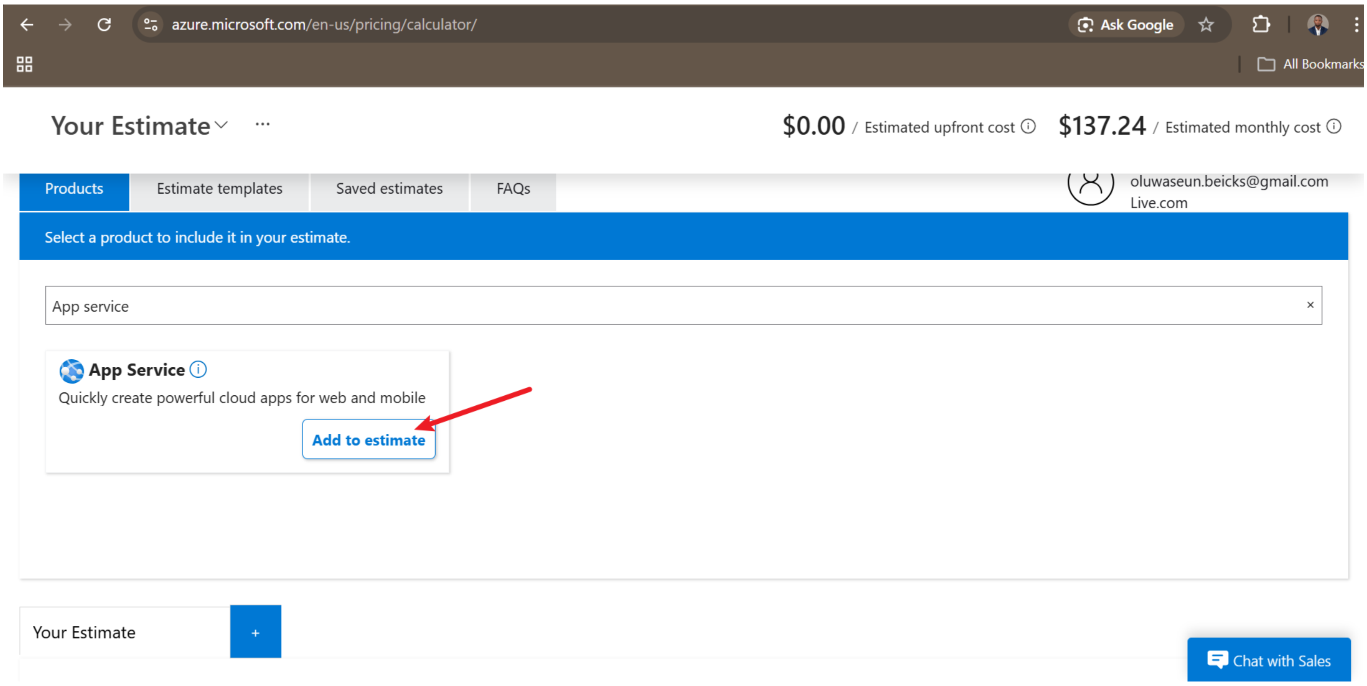


IaaS Estimated Monthly Cost: $137.24



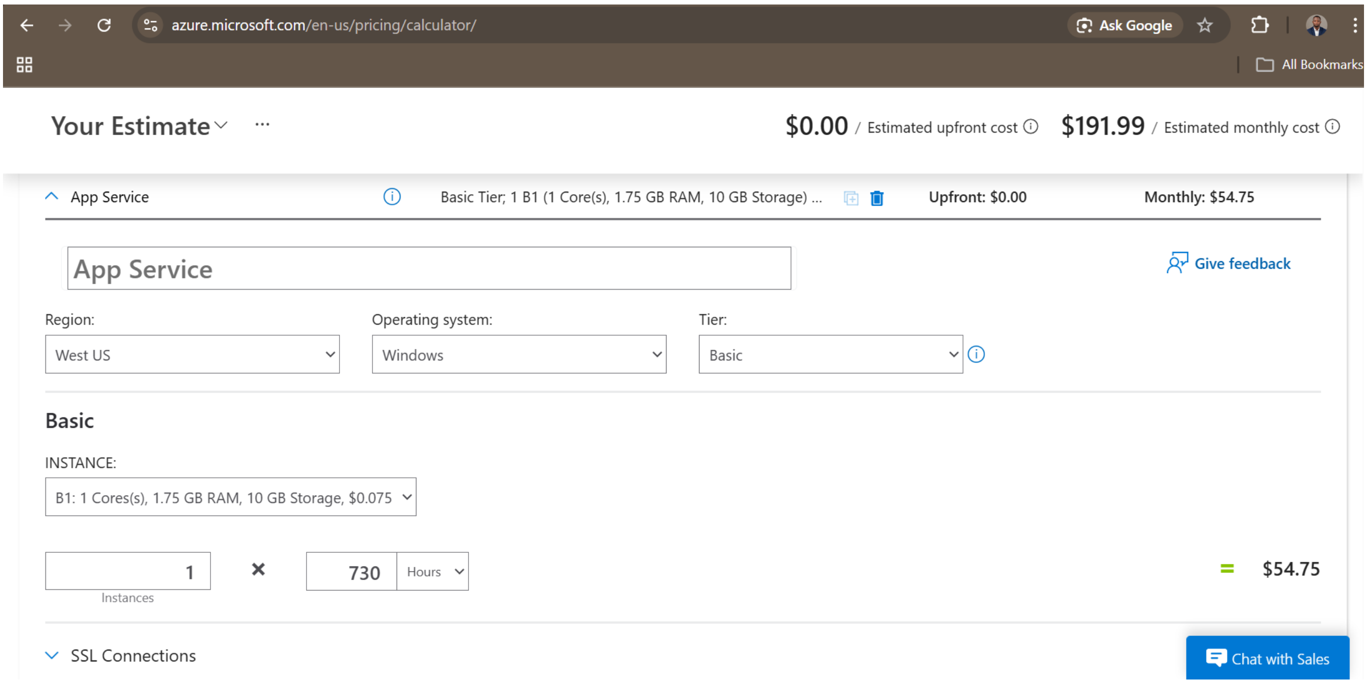
Step 3 — Estimate PaaS Costs (App Service)

Scrolled back to the search bar and added **App Service** to the estimate. PaaS abstracts infrastructure management — Azure handles OS patching, scaling, and availability.

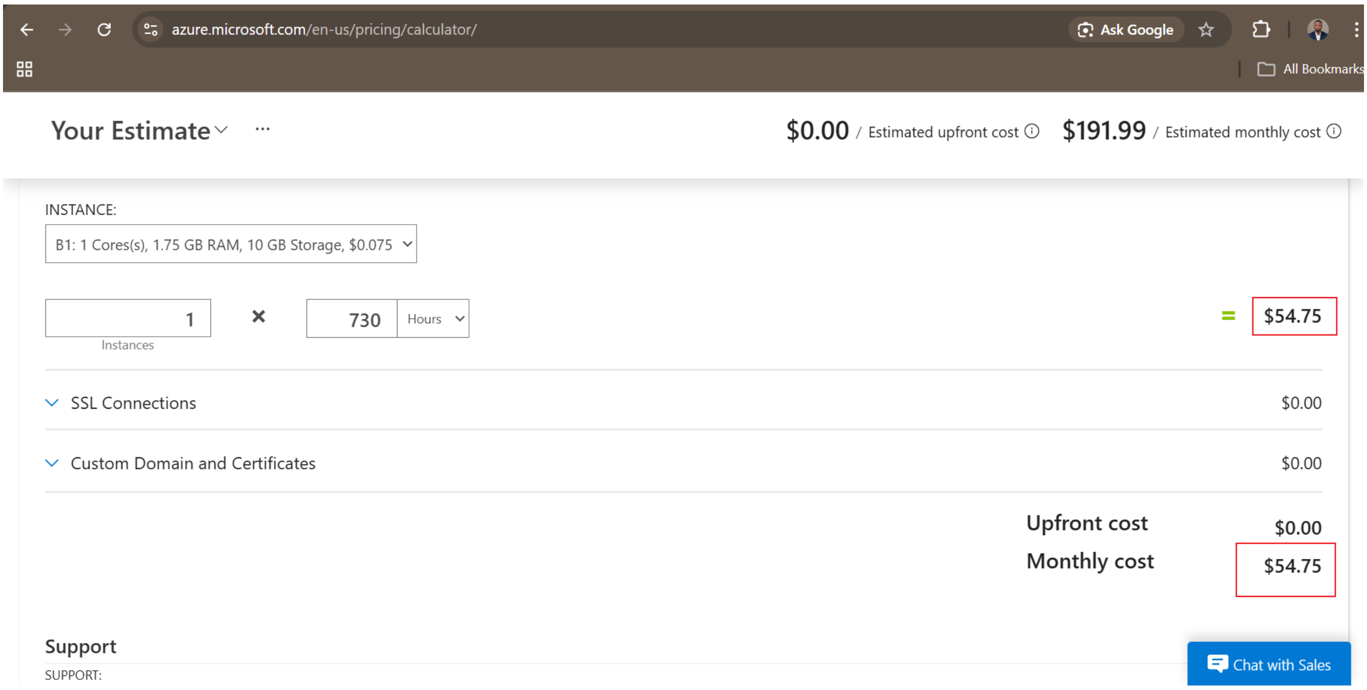


Configured App Service settings:

- **Region:** East US
- **Tier:** Standard
- **Instance:** B1 (1 Core, 1.75 GB RAM)
- **Hours:** 730 (full month)



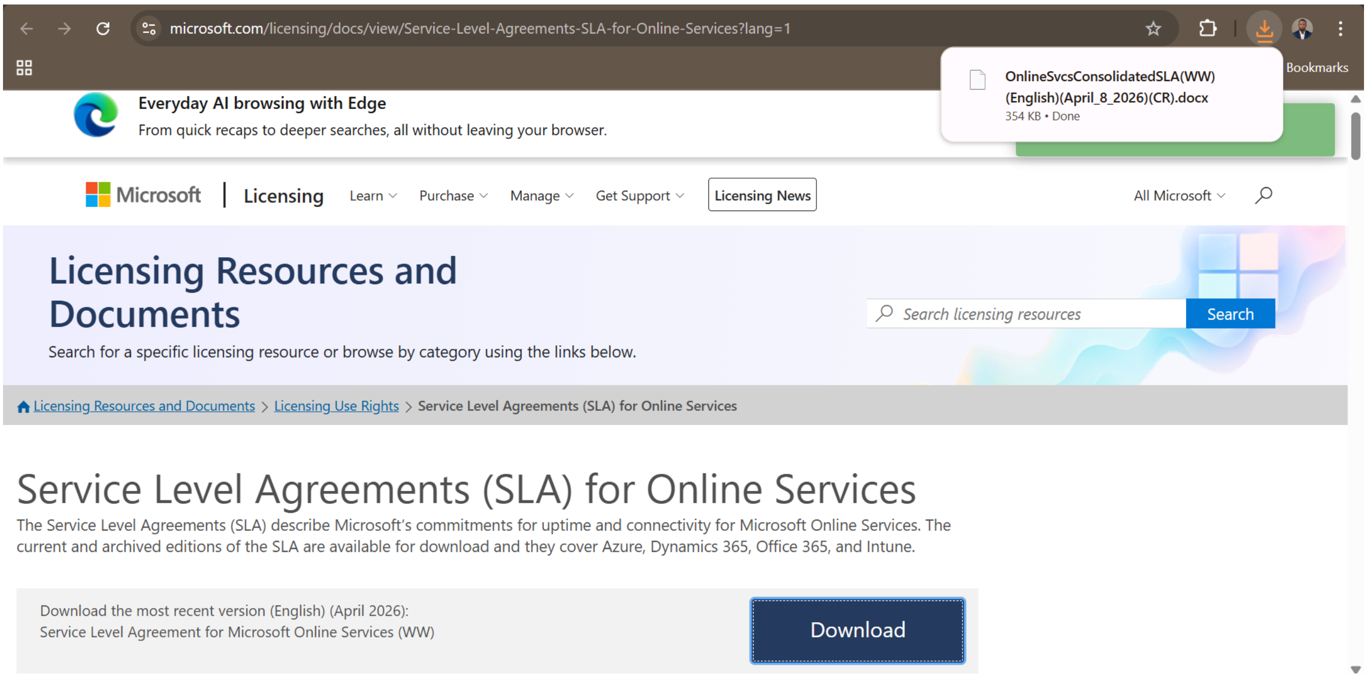
PaaS Estimated Monthly Cost: $54.75 — roughly 60% cheaper than the IaaS equivalent.



> **Key Insight:** With PaaS (App Service), CloudFirst Retail pays less, avoids OS management overhead, and gains automatic scaling — this is elasticity in action.

Step 4 — Explore Service Level Agreements (SLAs)

Opened the Azure SLA documentation page (`https://azure.microsoft.com/en-us/support/legal/sla/`) in a new tab and downloaded the **Service Level Agreement for Microsoft Online Services (WW)** Word document.



Opened the downloaded document and navigated to the Virtual Machines section.

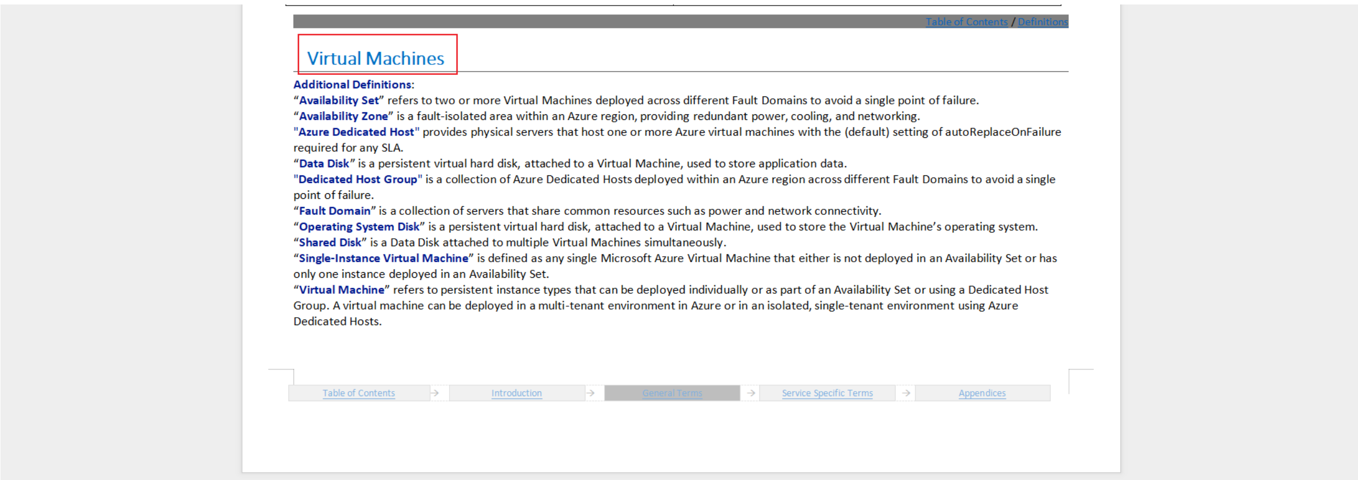


Table of Contents / Definitions

Virtual Machines

Additional Definitions:

"**Availability Set**" refers to two or more Virtual Machines deployed across different Fault Domains to avoid a single point of failure.
"**Availability Zone**" is a fault-isolated area within an Azure region, providing redundant power, cooling, and networking.
"**Azure Dedicated Host**" provides physical servers that host one or more Azure virtual machines with the (default) setting of autoReplaceOnFailure required for any SLA.
"**Data Disk**" is a persistent virtual hard disk, attached to a Virtual Machine, used to store application data.
"**Dedicated Host Group**" is a collection of Azure Dedicated Hosts deployed within an Azure region across different Fault Domains to avoid a single point of failure.
"**Fault Domain**" is a collection of servers that share common resources such as power and network connectivity.
"**Operating System Disk**" is a persistent virtual hard disk, attached to a Virtual Machine, used to store the Virtual Machine's operating system.
"**Shared Disk**" is a Data Disk attached to multiple Virtual Machines simultaneously.
"**Single-Instance Virtual Machine**" is defined as any single Microsoft Azure Virtual Machine that either is not deployed in an Availability Set or has only one instance deployed in an Availability Set.
"**Virtual Machine**" refers to persistent instance types that can be deployed individually or as part of an Availability Set or using a Dedicated Host Group. A virtual machine can be deployed in a multi-tenant environment in Azure or in an isolated, single-tenant environment using Azure Dedicated Hosts.

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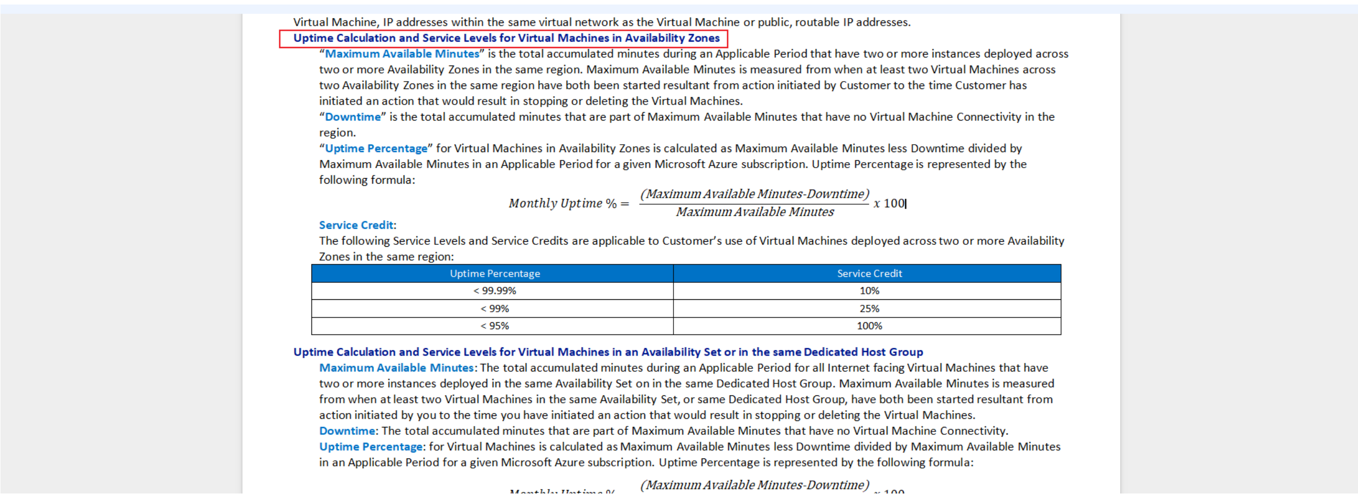
4a — VMs in Availability Zones: 99.99% SLA

Located the "Uptime Calculation and Service Levels for Virtual Machines in Availability Zones" table.



Virtual Machine, IP addresses within the same virtual network as the Virtual Machine or public, routable IP addresses.

Uptime Calculation and Service Levels for Virtual Machines in Availability Zones

"**Maximum Available Minutes**" is the total accumulated minutes during an Applicable Period that have two or more instances deployed across two or more Availability Zones in the same region. Maximum Available Minutes is measured from when at least two Virtual Machines across two Availability Zones in the same region have both been started resultant from action initiated by Customer to the time Customer has initiated an action that would result in stopping or deleting the Virtual Machines.

"**Downtime**" is the total accumulated minutes that are part of Maximum Available Minutes that have no Virtual Machine Connectivity in the region.

"**Uptime Percentage**" for Virtual Machines in Availability Zones is calculated as Maximum Available Minutes less Downtime divided by Maximum Available Minutes in an Applicable Period for a given Microsoft Azure subscription. Uptime Percentage is represented by the following formula:

$$Monthly\ Uptime\ \% = \frac{(Maximum\ Available\ Minutes\text{-}Downtime)}{Maximum\ Available\ Minutes} \times 100$$

Service Credit:

The following Service Levels and Service Credits are applicable to Customer's use of Virtual Machines deployed across two or more Availability Zones in the same region:

Uptime Percentage	Service Credit
< 99.99%	10%
< 99%	25%
< 95%	100%

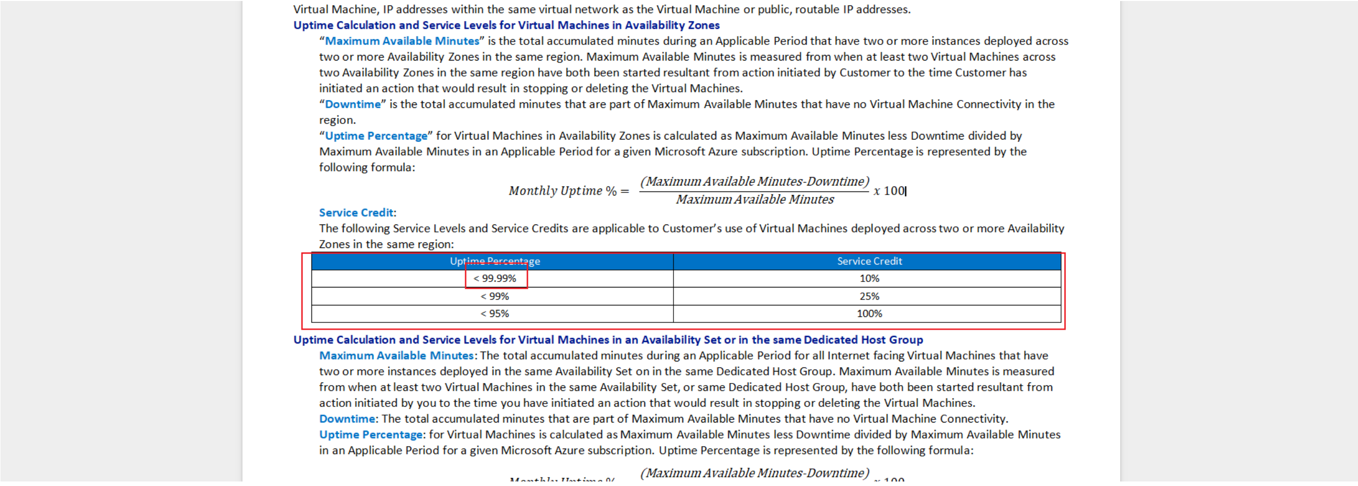
Uptime Calculation and Service Levels for Virtual Machines in an Availability Set or in the same Dedicated Host Group

Maximum Available Minutes: The total accumulated minutes during an Applicable Period for all Internet facing Virtual Machines that have two or more instances deployed in the same Availability Set or on in the same Dedicated Host Group. Maximum Available Minutes is measured from when at least two Virtual Machines in the same Availability Set, or same Dedicated Host Group, have both been started resultant from action initiated by you to the time you have initiated an action that would result in stopping or deleting the Virtual Machines.

Downtime: The total accumulated minutes that are part of Maximum Available Minutes that have no Virtual Machine Connectivity.

Uptime Percentage: for Virtual Machines is calculated as Maximum Available Minutes less Downtime divided by Maximum Available Minutes in an Applicable Period for a given Microsoft Azure subscription. Uptime Percentage is represented by the following formula:

(Maximum Available Minutes-Downtime)



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Service Credit:

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Uptime Percentage	Service Credit
< 99.99%	10%
< 99%	25%
< 95%	100%

Uptime Calculation and Service Levels for Virtual Machines in an Availability Set or in the same Dedicated Host Group

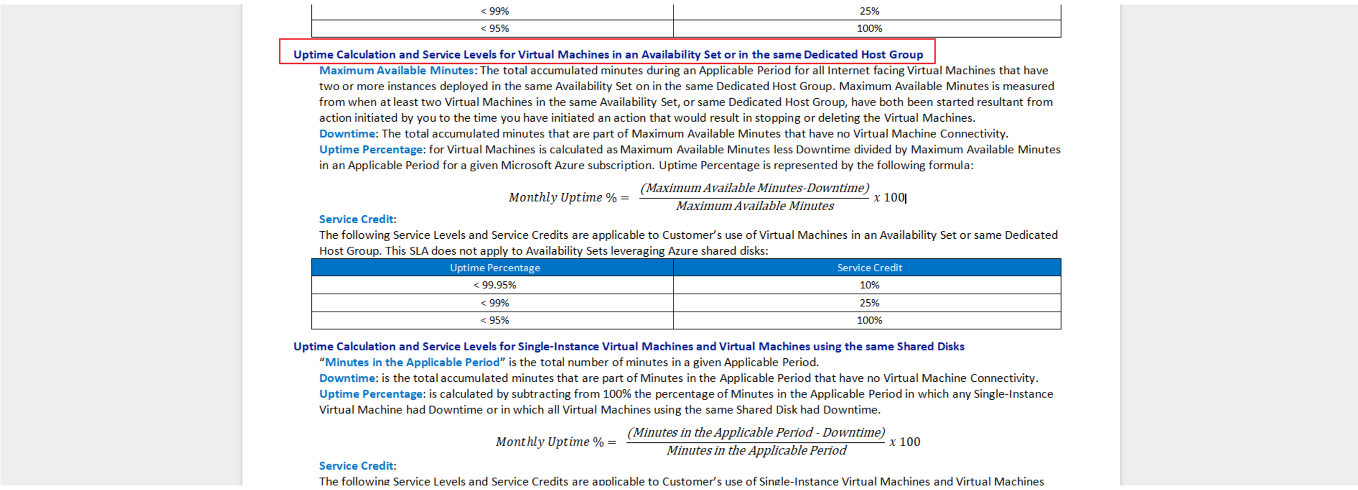
Maximum Available Minutes: The total accumulated minutes during an Applicable Period for all Internet facing Virtual Machines that have two or more instances deployed in the same Availability Set on in the same Dedicated Host Group. Maximum Available Minutes is measured from when at least two Virtual Machines in the same Availability Set, or same Dedicated Host Group, have both been started resultant from action initiated by you to the time you have initiated an action that would result in stopping or deleting the Virtual Machines.

Downtime: The total accumulated minutes that are part of Maximum Available Minutes that have no Virtual Machine Connectivity.

Uptime Percentage: for Virtual Machines is calculated as Maximum Available Minutes less Downtime divided by Maximum Available Minutes in an Applicable Period for a given Microsoft Azure subscription. Uptime Percentage is represented by the following formula:

4b — VMs in Availability Set: 99.95% SLA

Located the "Uptime Calculation and Service Levels for Virtual Machines in an Availability Set" table.



< 99%	25%
< 95%	100%

Uptime Calculation and Service Levels for Virtual Machines in an Availability Set or in the same Dedicated Host Group

Maximum Available Minutes: The total accumulated minutes during an Applicable Period for all Internet facing Virtual Machines that have two or more instances deployed in the same Availability Set on in the same Dedicated Host Group. Maximum Available Minutes is measured from when at least two Virtual Machines in the same Availability Set, or same Dedicated Host Group, have both been started resultant from action initiated by you to the time you have initiated an action that would result in stopping or deleting the Virtual Machines.

Downtime: The total accumulated minutes that are part of Maximum Available Minutes that have no Virtual Machine Connectivity.

Uptime Percentage: for Virtual Machines is calculated as Maximum Available Minutes less Downtime divided by Maximum Available Minutes in an Applicable Period for a given Microsoft Azure subscription. Uptime Percentage is represented by the following formula:

$$Monthly\ Uptime\ \% = \frac{(Maximum\ Available\ Minutes\text{-}Downtime)}{Maximum\ Available\ Minutes} \times 100$$

Service Credit:

The following Service Levels and Service Credits are applicable to Customer's use of Virtual Machines in an Availability Set or same Dedicated Host Group. This SLA does not apply to Availability Sets leveraging Azure shared disks:

Uptime Percentage	Service Credit
< 99.95%	10%
< 99%	25%
< 95%	100%

Uptime Calculation and Service Levels for Single-Instance Virtual Machines and Virtual Machines using the same Shared Disks

"**Minutes in the Applicable Period**" is the total number of minutes in a given Applicable Period.

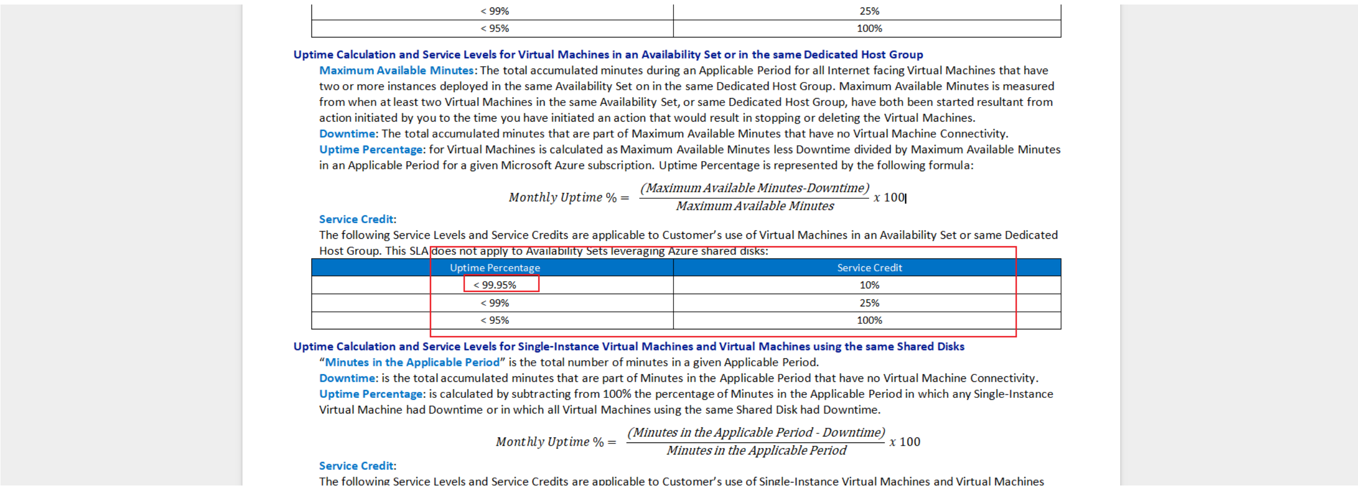
Downtime: is the total accumulated minutes that are part of Minutes in the Applicable Period that have no Virtual Machine Connectivity.

Uptime Percentage: is calculated by subtracting from 100% the percentage of Minutes in the Applicable Period in which any Single-Instance Virtual Machine had Downtime or in which all Virtual Machines using the same Shared Disk had Downtime.

$$Monthly\ Uptime\ \% = \frac{(Minutes\ in\ the\ Applicable\ Period\ \text{-}\ Downtime)}{Minutes\ in\ the\ Applicable\ Period} \times 100$$

Service Credit:

The following Service Levels and Service Credits are applicable to Customer's use of Single-Instance Virtual Machines and Virtual Machines



< 99%	25%
< 95%	100%

Uptime Calculation and Service Levels for Virtual Machines in an Availability Set or in the same Dedicated Host Group

Maximum Available Minutes: The total accumulated minutes during an Applicable Period for all Internet facing Virtual Machines that have two or more instances deployed in the same Availability Set on in the same Dedicated Host Group. Maximum Available Minutes is measured from when at least two Virtual Machines in the same Availability Set, or same Dedicated Host Group, have both been started resultant from action initiated by you to the time you have initiated an action that would result in stopping or deleting the Virtual Machines.

Downtime: The total accumulated minutes that are part of Maximum Available Minutes that have no Virtual Machine Connectivity.

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Service Credit:

The following Service Levels and Service Credits are applicable to Customer's use of Virtual Machines in an Availability Set or same Dedicated Host Group. This SLA does not apply to Availability Sets leveraging Azure shared disks:

Uptime Percentage	Service Credit
< 99.95%	10%
< 99%	25%
< 95%	100%

Uptime Calculation and Service Levels for Single-Instance Virtual Machines and Virtual Machines using the same Shared Disks

"**Minutes in the Applicable Period**" is the total number of minutes in a given Applicable Period.

Downtime: is the total accumulated minutes that are part of Minutes in the Applicable Period that have no Virtual Machine Connectivity.

Uptime Percentage: is calculated by subtracting from 100% the percentage of Minutes in the Applicable Period in which any Single-Instance Virtual Machine had Downtime or in which all Virtual Machines using the same Shared Disk had Downtime.

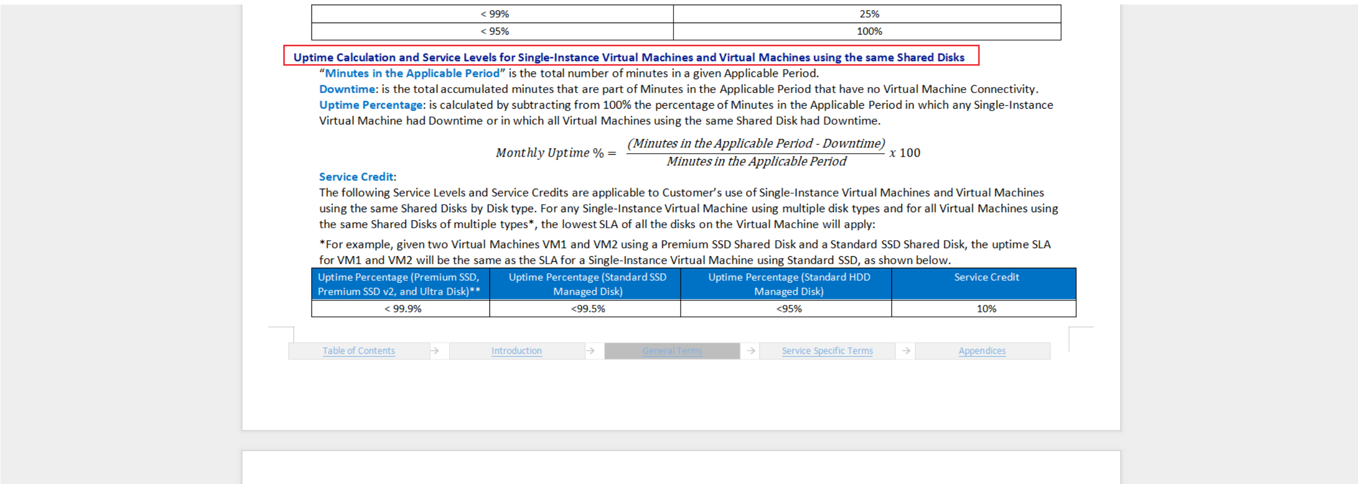
$$Monthly\ Uptime\ \% = \frac{(Minutes\ in\ the\ Applicable\ Period\ \text{-}\ Downtime)}{Minutes\ in\ the\ Applicable\ Period} \times 100$$

Service Credit:

The following Service Levels and Service Credits are applicable to Customer's use of Single-Instance Virtual Machines and Virtual Machines

4c — Single-Instance VM (Premium SSD): 99.9% SLA

Located the "Uptime Calculation and Service Levels for Single-Instance Virtual Machines" table and reviewed the Premium SSD column.



< 99%	25%
< 95%	100%

Uptime Calculation and Service Levels for Single-Instance Virtual Machines and Virtual Machines using the same Shared Disks

"**Minutes in the Applicable Period**" is the total number of minutes in a given Applicable Period.
Downtime: is the total accumulated minutes that are part of Minutes in the Applicable Period that have no Virtual Machine Connectivity.
Uptime Percentage: is calculated by subtracting from 100% the percentage of Minutes in the Applicable Period in which any Single-Instance Virtual Machine had Downtime or in which all Virtual Machines using the same Shared Disk had Downtime.

$$\textit{Monthly Uptime \%} = \frac{(\textit{Minutes in the Applicable Period} - \textit{Downtime})}{\textit{Minutes in the Applicable Period}} \times 100$$

Service Credit:
The following Service Levels and Service Credits are applicable to Customer's use of Single-Instance Virtual Machines and Virtual Machines using the same Shared Disks by Disk type. For any Single-Instance Virtual Machine using multiple disk types and for all Virtual Machines using the same Shared Disks of multiple types*, the lowest SLA of all the disks on the Virtual Machine will apply:
*For example, given two Virtual Machines VM1 and VM2 using a Premium SSD Shared Disk and a Standard SSD Shared Disk, the uptime SLA for VM1 and VM2 will be the same as the SLA for a Single-Instance Virtual Machine using Standard SSD, as shown below.

Uptime Percentage (Premium SSD, Premium SSD v2, and Ultra Disk)**	Uptime Percentage (Standard SSD Managed Disk)	Uptime Percentage (Standard HDD Managed Disk)	Service Credit
< 99.9%	<99.5%	<95%	10%

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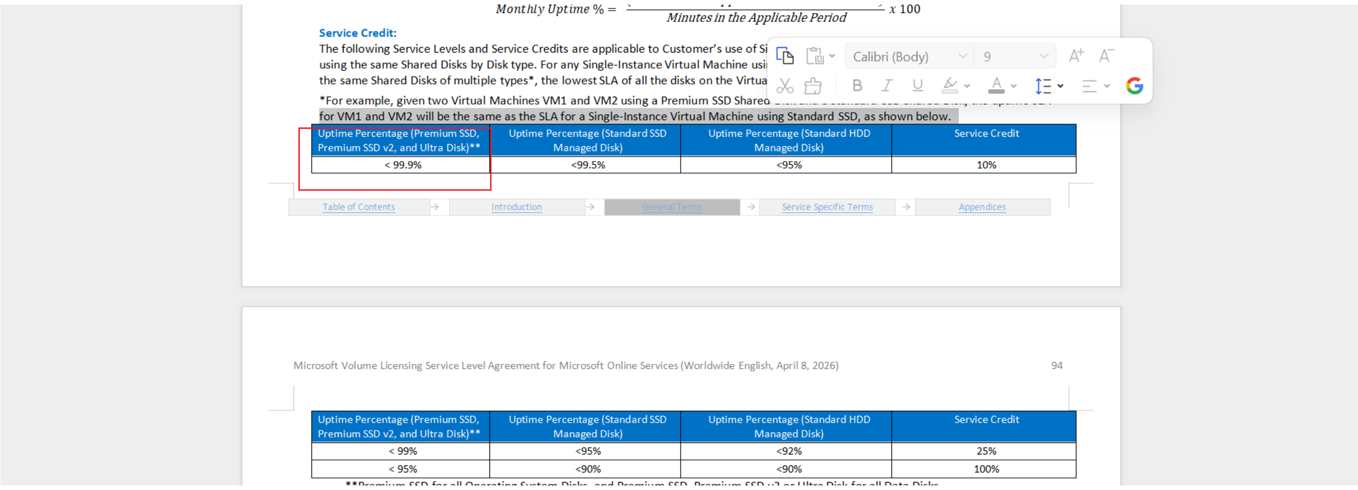

$$\textit{Monthly Uptime \%} = \frac{\ldots}{\textit{Minutes in the Applicable Period}} \times 100$$

Service Credit:
The following Service Levels and Service Credits are applicable to Customer's use of Si
using the same Shared Disks by Disk type. For any Single-Instance Virtual Machine usi
the same Shared Disks of multiple types*, the lowest SLA of all the disks on the Virtua
*For example, given two Virtual Machines VM1 and VM2 using a Premium SSD Shared
for VM1 and VM2 will be the same as the SLA for a Single-Instance Virtual Machine using Standard SSD, as shown below.

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Uptime Percentage (Premium SSD, Premium SSD v2, and Ultra Disk)**	Uptime Percentage (Standard SSD Managed Disk)	Uptime Percentage (Standard HDD Managed Disk)	Service Credit
< 99.9%	<99.5%	<95%	10%

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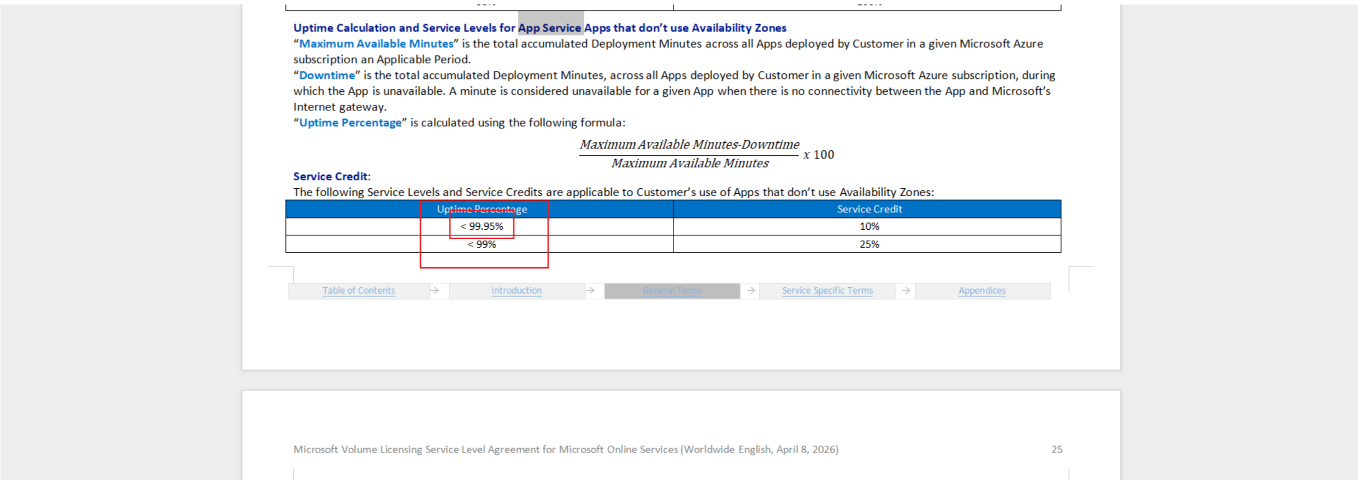
Uptime Percentage (Premium SSD, Premium SSD v2, and Ultra Disk)**	Uptime Percentage (Standard SSD Managed Disk)	Uptime Percentage (Standard HDD Managed Disk)	Service Credit
< 99%	<95%	<92%	25%
< 95%	<90%	<90%	100%

4d — App Service SLA: 99.95%

Used Ctrl+F to locate the App Service SLA section and noted the uptime guarantee.



Uptime Calculation and Service Levels for App Service Apps that don't use Availability Zones
"**Maximum Available Minutes**" is the total accumulated Deployment Minutes across all Apps deployed by Customer in a given Microsoft Azure subscription an Applicable Period.
"**Downtime**" is the total accumulated Deployment Minutes, across all Apps deployed by Customer in a given Microsoft Azure subscription, during which the App is unavailable. A minute is considered unavailable for a given App when there is no connectivity between the App and Microsoft's Internet gateway.
"**Uptime Percentage**" is calculated using the following formula:

$$\frac{\textit{Maximum Available Minutes-Downtime}}{\textit{Maximum Available Minutes}} \times 100$$

Service Credit:
The following Service Levels and Service Credits are applicable to Customer's use of Apps that don't use Availability Zones:

Uptime Percentage	Service Credit
< 99.95%	10%
< 99%	25%

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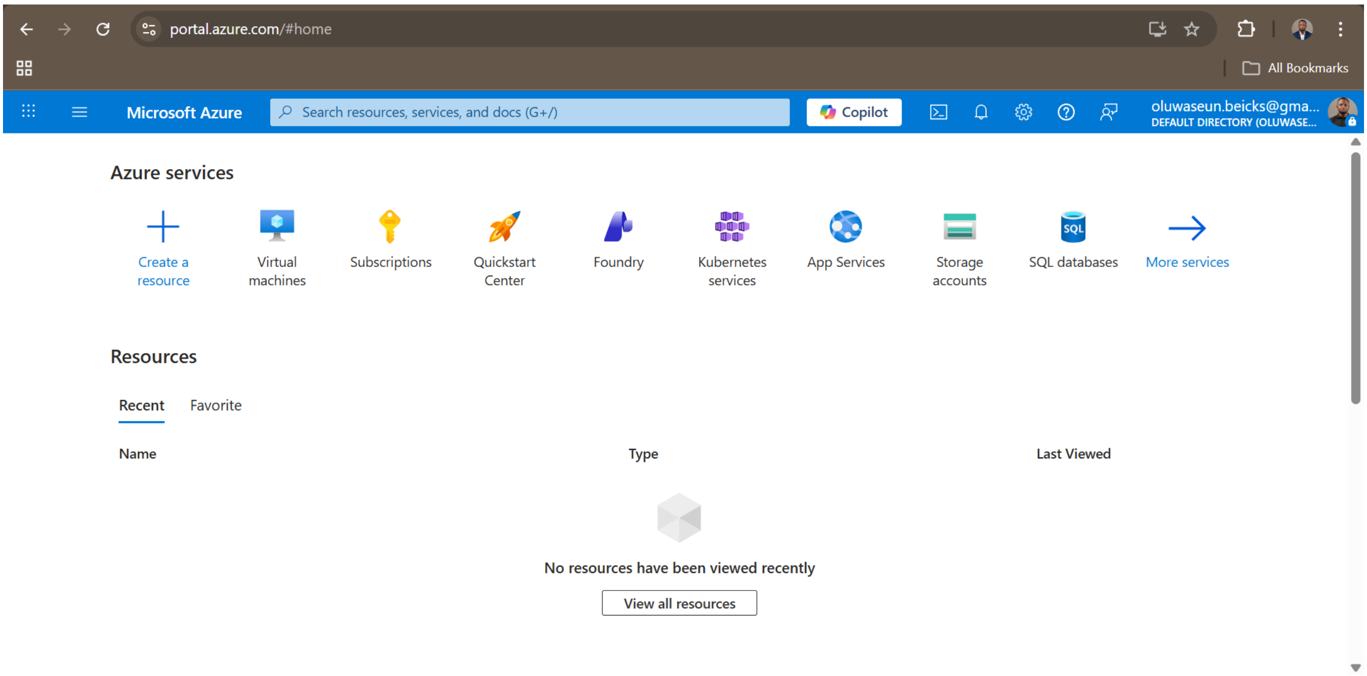
> **Composite SLA Concept:** App Service (99.95%) × SQL Database (99.99%) = **~99.94% composite SLA**. More dependencies always reduce overall availability — composite SLAs multiply, they don't average.

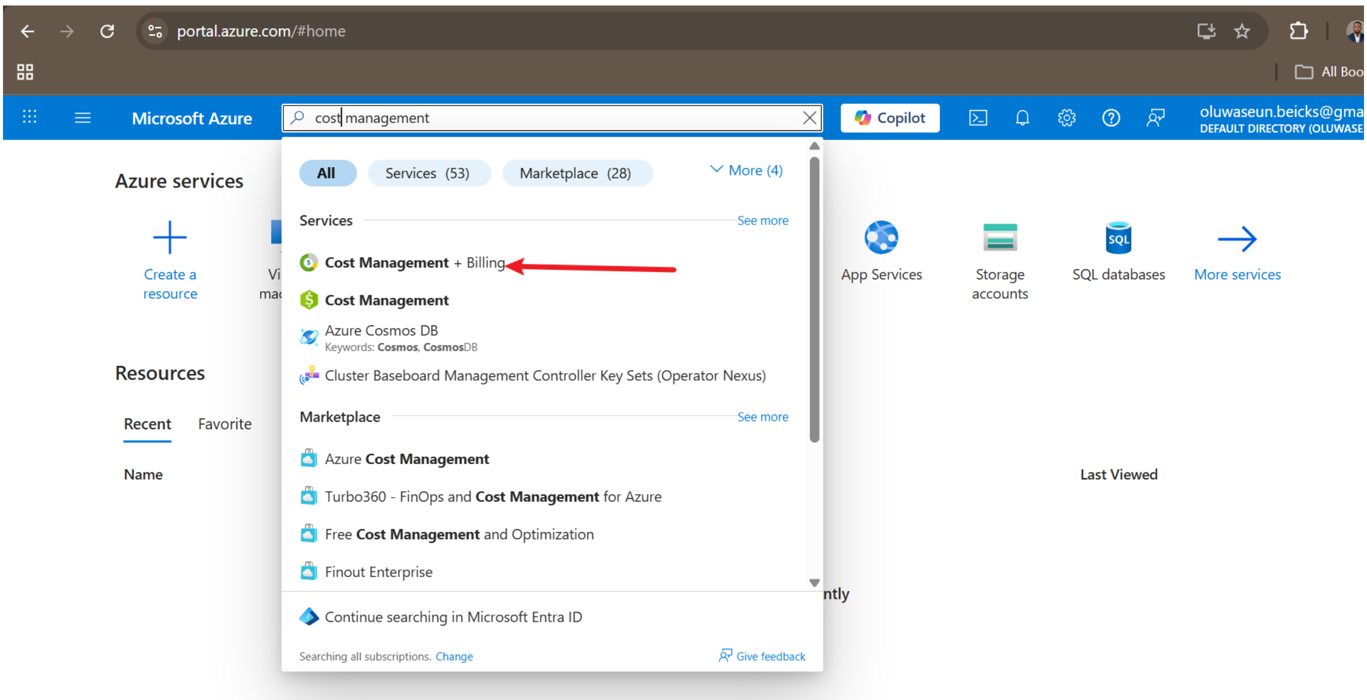
Configuration	SLA
VMs across Availability Zones	99.99%
VMs in Availability Set	99.95%

Configuration	SLA
Single-Instance VM (Premium SSD)	99.9%
App Service	99.95%

Step 5 — Access Azure Cost Management + Billing

Signed into the Azure Portal (`https://portal.azure.com`) and searched for **Cost Management + Billing**.

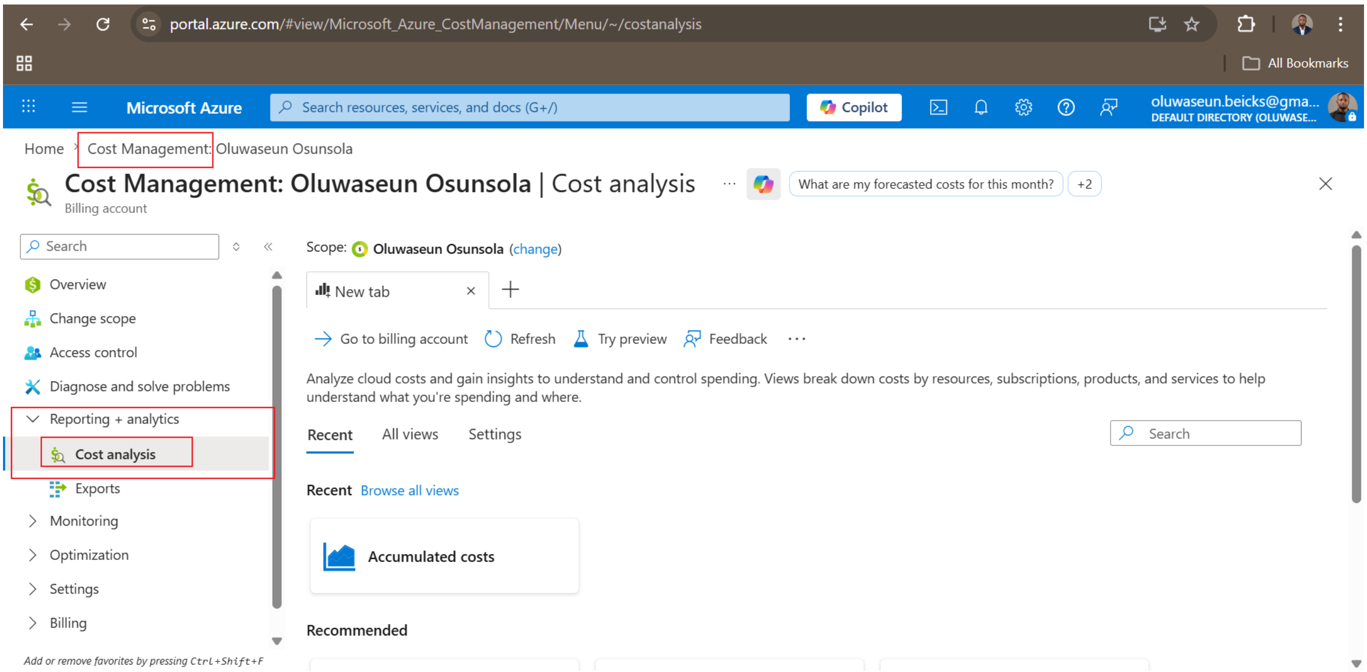




Step 6 — Explore Cost Management Features

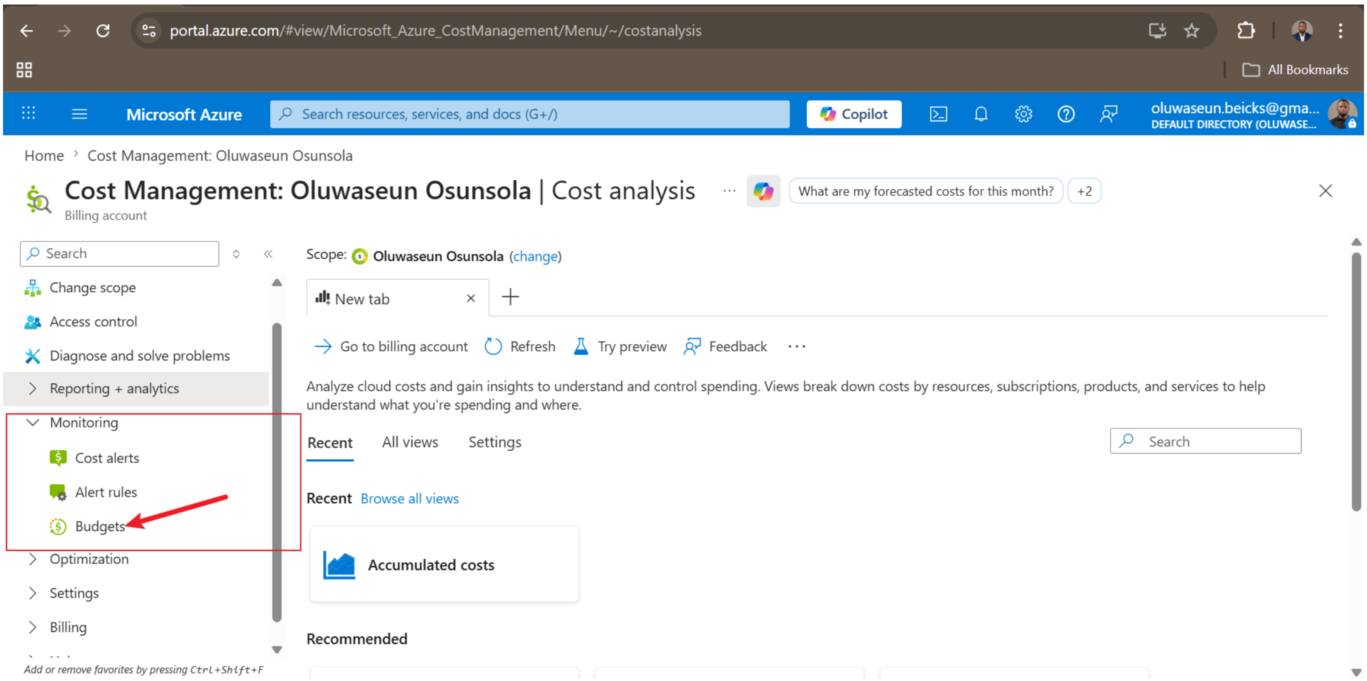
Cost Analysis Dashboard

Clicked **Cost analysis** under Cost Management → Reporting + Analytics to view the spending dashboard. This is where consumption-based pricing becomes tangible — every resource charge is visible in real time.

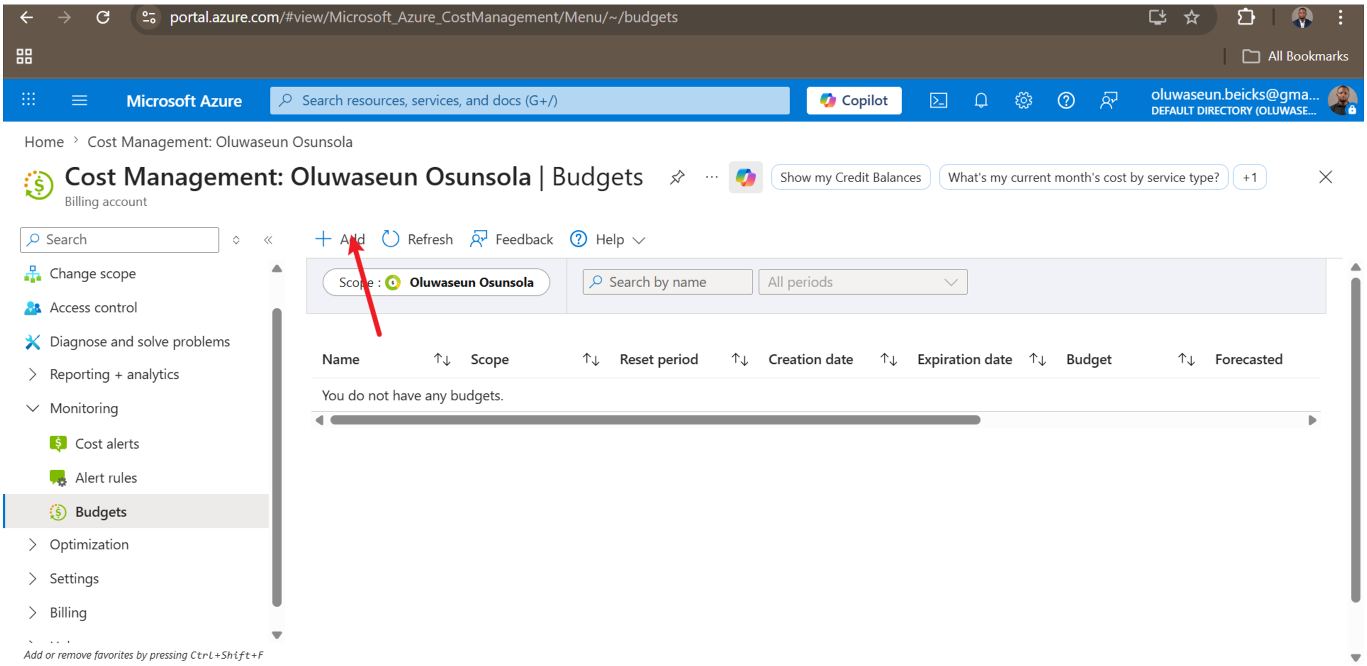


Budgets

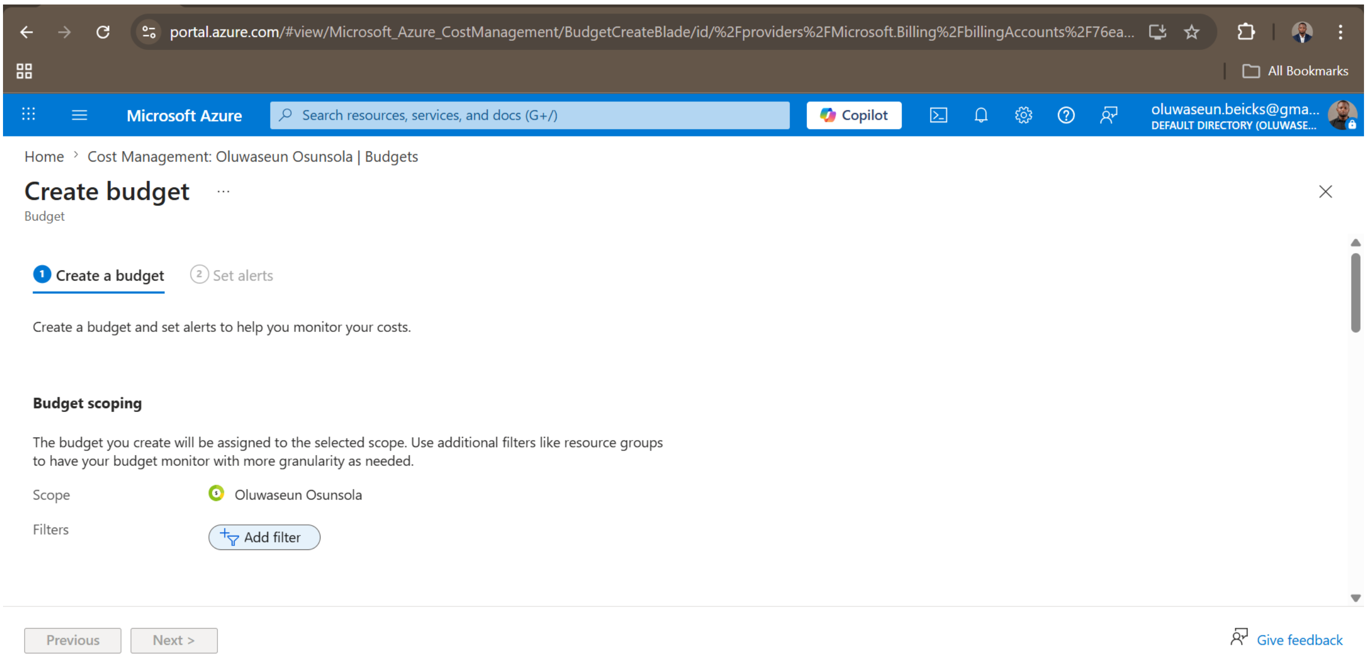
Navigated to **Budgets** under the Monitoring section. This is where spending limits and threshold alerts are configured.



Clicked **Add** to begin creating a budget.

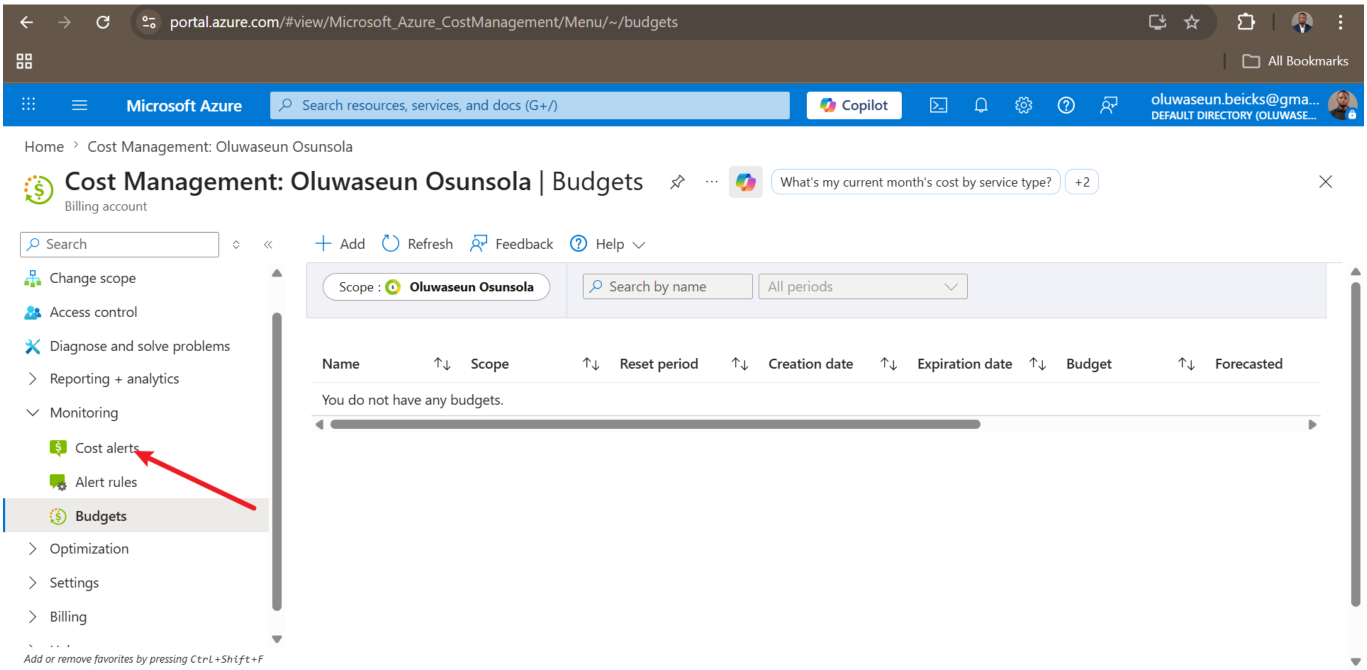


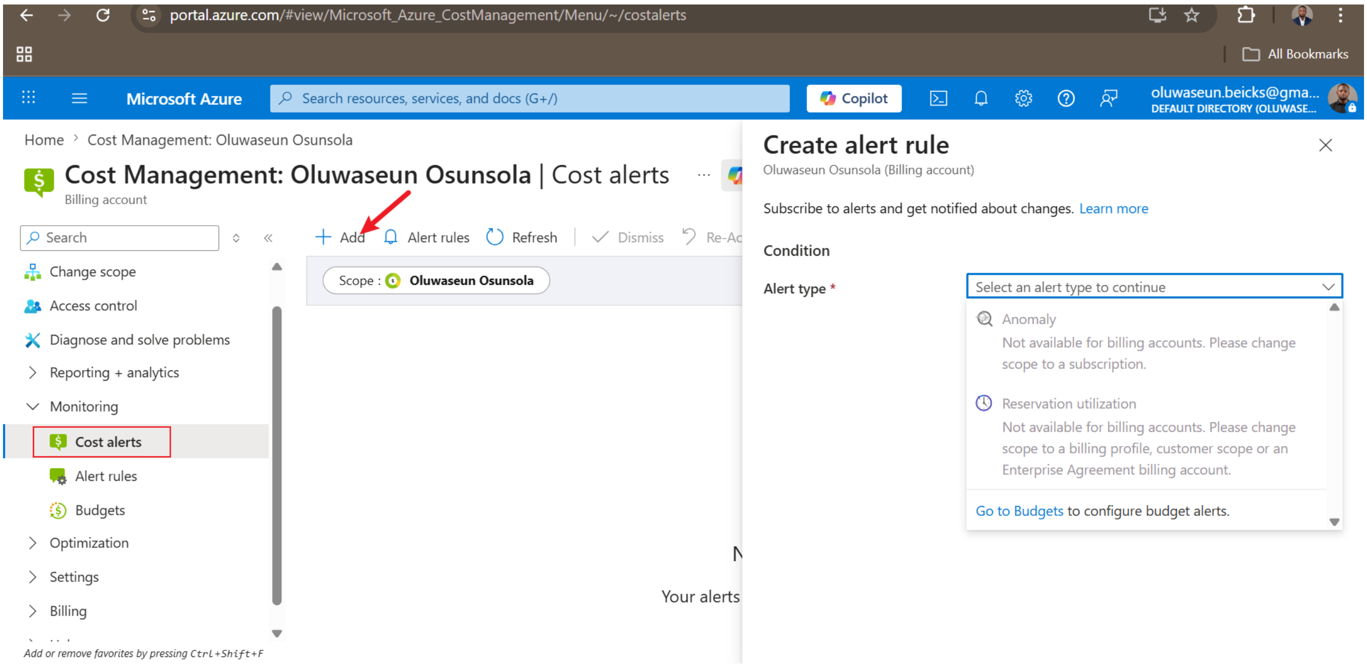
Reviewed the budget creation form — teams can set monthly limits, define alert thresholds (e.g., 80%, 90%), and configure automatic notifications.



Cost Alerts

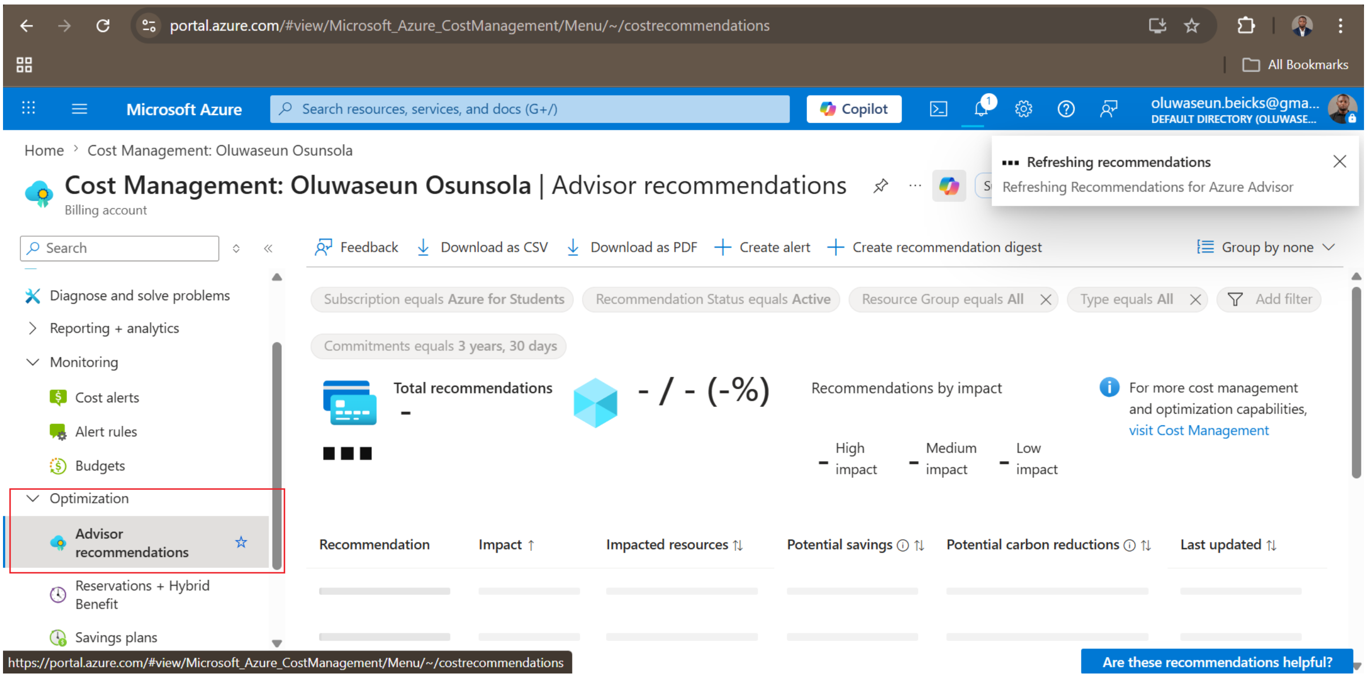
Navigated back to **Cost alerts** to review alert configuration options for notifying teams when spending approaches defined thresholds.





Advisor Recommendations

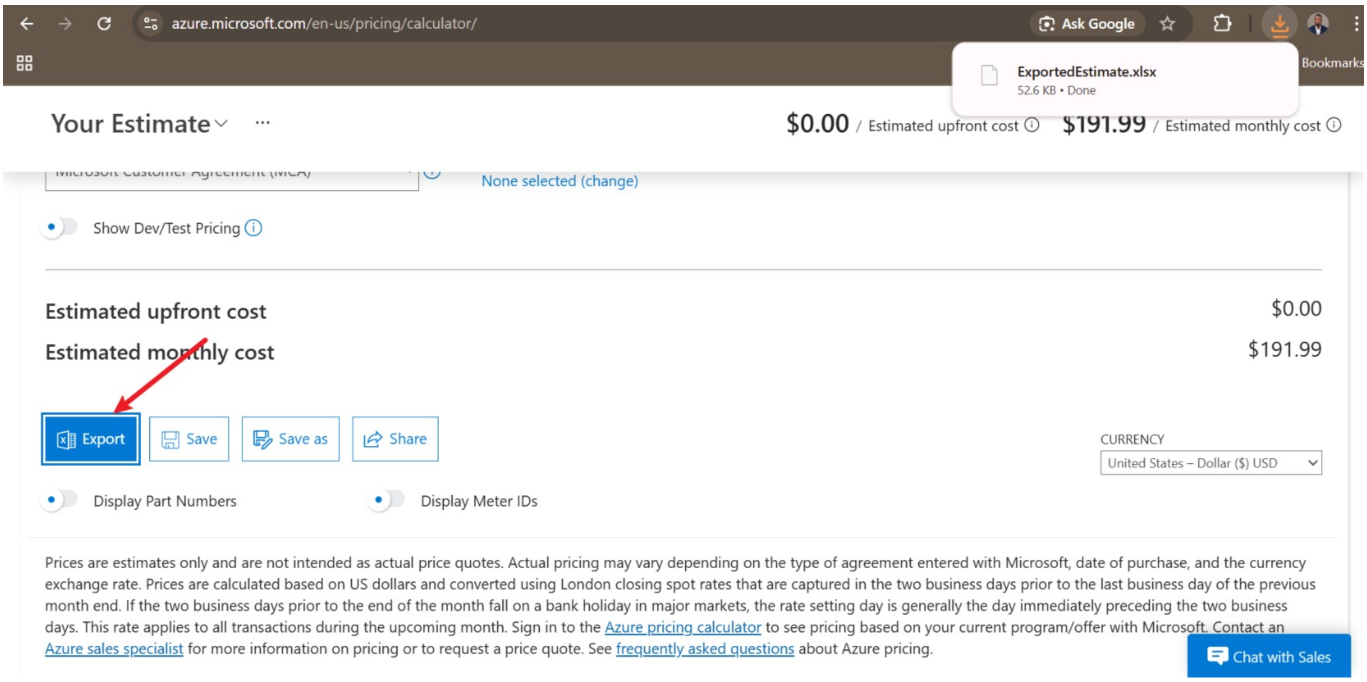
Clicked **Advisor recommendations** under Optimization to review AI-powered cost reduction suggestions. On a free account with no deployed resources, this section may show no recommendations yet.



> **Why This Matters:** These tools transform cloud spending from unpredictable to manageable. Budgets + alerts + Advisor recommendations give full visibility and control over Azure's OpEx model.

Step 7 & 8 — Review Shared Responsibility & Export Cost Estimate

Returned to the Azure Pricing Calculator to review the full estimate and apply Shared Responsibility thinking to each service model. Then exported the estimate as an Excel (`.xlsx`) file for stakeholder presentation.



Shared Responsibility Summary

Responsibility Area	IaaS (VM)	PaaS (App Service)	SaaS
Physical datacenter & hosts	Microsoft	Microsoft	Microsoft
Operating system patching	**Customer**	Microsoft	Microsoft

Responsibility Area	IaaS (VM)	PaaS (App Service)	SaaS
Network security group rules	**Customer**	Microsoft	Microsoft
Application security & code	**Customer**	**Customer**	**Customer**
Data security & encryption	**Customer**	**Customer**	**Customer**
Identity & access management	**Customer**	**Customer**	**Customer**

> **Rule of thumb:** No matter the service model, **data security, identity/access management, and endpoint protection are always the customer's responsibility**.

📊 Cost Comparison Summary

Service	Model	Monthly Estimate	OS Patching	Scaling
2× D2s v3 Virtual Machines	IaaS	**$137.24**	Customer	Manual
App Service S1	PaaS	**$54.75**	Microsoft	Automatic

PaaS delivers ~60% cost savings while reducing operational overhead and offering built-in elasticity — a compelling business case for CloudFirst Retail.

📝 Key Takeaways

Service Model Selection

- **IaaS** → Maximum control, maximum responsibility, higher costs. Best for legacy apps or OS-level customization.
- **PaaS** → Managed platform, automatic updates, lower costs. Best when the focus is application code, not infrastructure.
- **SaaS** → Zero infrastructure management, pure consumption (e.g., Microsoft 365).

Consumption-Based Pricing (OpEx)

- Pay only for what you use — no upfront capital expenditure (CapEx)
- Cost Management tools (budgets, alerts, Advisor) are essential for cost predictability
- Reserved Instances can save up to 72% for predictable, long-running workloads

SLA Mechanics

- More "nines" = less downtime: 99.99% = ~52 minutes/year of allowed downtime
- Composite SLAs are calculated by multiplying individual SLAs — dependencies always reduce overall availability
- Use Availability Zones to achieve the highest VM SLA tier (99.99%)

🧪 Exam Tips (AZ-900)

Question Pattern	Answer
Most OS-level control?	IaaS — Virtual Machines
Least infrastructure management?	SaaS — Microsoft 365
Who patches Azure SQL Database?	Microsoft (PaaS platform management)
VM compromised via open RDP port — who's responsible?	Customer (network config is customer responsibility)
Consumption-based pricing = which financial model?	OpEx (Operational Expenditure)
App Service (99.95%) + SQL DB (99.99%) composite SLA?	~99.94%
SLA with < 1 hour downtime per year?	99.99% (52.56 min/year)

🔗 Resources

- Azure Pricing Calculator
- Azure SLA Documentation
- Azure Portal — Cost Management
- Dion Training — AZ-900 Course

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*Lab completed by **Oluwaseun Osunsola** as part of the Microsoft Azure Fundamentals (AZ-900) certification preparation — Section 2: Cloud Computing Concepts.*